Real-Time SAR Ship Detection:
A Speckle-Model-Based DSP Pipeline

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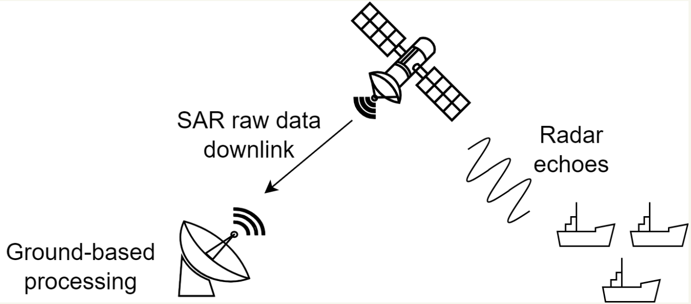
Outline

- Introduction
- Problem and Motivation
- Highlight of Contribution
- System Definition
- Previous Methods
- Proposed Method
- Comparison
- Conclusion

Introduction

- SAR raw data are downlinked for ground-based processing, including SAR focusing, image enhancement, and ship detection.
- Detected ships provide information for maritime surveillance.



Problem and Motivation

- Ground-Based Processing
  - Large raw-data volume
  - Limited downlink bandwidth
  - Delayed target information
- Onboard Processing
  - Reduced data transmission and detection latency
  - Challenge: Limited computation, memory, and power
- Research Goal
  - Lightweight SAR processing for low-latency ship detection

Evaluation Framework

- Ray tracing for SAR echo generation
- Enhanced SAR imaging as the primary research focus
- Onboard YOLO-based ship detection for performance evaluation (YOLO: You Only Look Once)